

# Parag Jain

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## Education

### University of Edinburgh

Edinburgh, UK

CDT-NLP PhD, Natural Language Processing, Advisor: [Prof. Mirella Lapata](#)

Thesis: Explicit context representations for conversational language understanding

Fully funded by Huawei and UKRI.

### Indian Institute of Technology

Hyderabad, India

M.Tech, Computer Science and Engineering, Advisor: [Prof. Vineeth N Balasubramanian](#)

Thesis: Metric Learning for Clustering in Streaming Large-Scale Data

Worked with [Dr. TVS Udaya Bhaskar](#) (INCOIS, India) on Ocean Biome Classification

Awarded A+ (highest possible) for research thesis

### Jabalpur Engineering College

Jabalpur, India

B.E., Computer Science and Engineering

## Experience

### AI Research Scientist

Feb 2025 - Current

Meta Superintelligence Labs

### Research Assistant

Feb 2024 - Dec 2024

University of Edinburgh

### Google DeepMind

July 2023 - Nov 2023

Student Researcher

### Tutor, University of Edinburgh

Jan 2021 - Apr 2021

Natural Language Understanding, Generation, and Machine Translation

### IBM Research

Aug 2016 - Sept 2019

Research Engineer

### Amazon

July 2015 - Aug 2016

Software Development Engineer

### Teaching Assistant, IIT Hyderabad

July 2013 - June 2015

Numerical Linear Algebra, Database Management Systems, and Advanced Compiler Design

## Conference Tutorial

### Structured Data Summarization

Delivered a 3-hour tutorial at the Annual Meeting of the Association for Computational Linguistics (ACL), 2019, Florence (Italy).

## Journal Publications

### Memory-Based Semantic Parsing

**Parag Jain**, Mirella Lapata

Transactions of the Association for Computational Linguistics

2021

### Scalable Micro-planned Generation of Discourse from Structured Data

Anirban Laha\*, **Parag Jain**\*, Abhijit Mishra\*, Karthik Sankaranarayanan

Computational Linguistics Journal

2019

Used for real-time verbal match insights during the spectator-free U.S. Open 2020, held without fans due to COVID-19.

In news: [\[US Open Blog\]](#) [\[Times of India\]](#) [\[Digital Creed\]](#)

## Other Publications

### Don't Waste Mistakes: Leveraging Negative RL-Groups via Confidence Reweighting

Yunzhen Feng, **Parag Jain**, Anthony Hartshorn, Julia Kempe, Yaqi Duan

Under Review at International Conference on Learning Representations (ICLR)

2025

### What Characterizes Effective Reasoning? Revisiting Length, Review, and Structure of CoT

Yunzhen Feng, Julia Kempe, Cheng Zhang, **Parag Jain**, Anthony Hartshorn

Under Review at International Conference on Learning Representations (ICLR)

2025

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| <b>Compute as Teacher: Turning Inference Compute Into Reference-Free Supervision</b><br>Dulhan Jayalath, Shashwat Goel, Thomas Foster, <b>Parag Jain</b> , Suchin Gururangan, Cheng Zhang, Anirudh Goyal, Alan Schelten<br>Under Review at International Conference on Learning Representations (ICLR)                                                              | 2025 |
| <b>The llama 4 herd: The beginning of a new era of natively multimodal AI innovation</b><br>Meta AI                                                                                                                                                                                                                                                                 | 2025 |
| <b>Integrating Large Language Models with Graph-based Reasoning for Conversational Question Answering</b><br><b>Parag Jain</b> , Mirella Lapata<br>arXiv:2407.09506                                                                                                                                                                                                 | 2024 |
| <b>Structsum Generation for Faster Text Comprehension</b><br><b>Parag Jain</b> , Andreea Marzoca, Francesco Piccinno<br>Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)                                                                                                                                               | 2024 |
| <b>Semantic Parsing for Conversational Question Answering over Knowledge Graphs</b><br>Laura Perez-Beltrachini, <b>Parag Jain</b> , Emilio Monti, Mirella Lapata<br>Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)                                                                              | 2023 |
| <b>Conversational Semantic Parsing using Dynamic Context Graphs</b><br><b>Parag Jain</b> , Mirella Lapata<br>Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)                                                                                                                                                         | 2023 |
| <b>Multi-Document Summarization with Centroid-Based Pretraining</b><br>Ratish Puduppully, <b>Parag Jain</b> , Nancy F. Chen, Mark Steedman<br>Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)                                                                                                                         | 2022 |
| <b>Bootstrapping chatbot interfaces to databases</b><br>Ashish Mittal, Diptikalyan Saha, <b>Parag Jain</b> , Jaydeep Sen, Manasa Jammi, Karthik Sankaranarayanan<br>Proceedings of the 3 <sup>rd</sup> ACM India Joint International Conference on Data Science and Management of Data                                                                              | 2021 |
| <b>Unsupervised Controllable Text Formalization</b><br><b>Parag Jain</b> , Abhijit Mishra, Amar Prakash Azad, Karthik Sankaranarayanan<br>Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)                                                                                                                                                      | 2019 |
| <b>Storytelling from structured data and knowledge graphs: An NLG perspective</b><br>Abhijit Mishra, Anirban Laha, Karthik Sankaranarayanan, <b>Parag Jain</b> , Saravanan Krishnan<br>Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL, Tutorial Abstracts)                                                            | 2019 |
| <b>Unified Semantic Parsing with Weak Supervision</b><br>Priyanka Agrawal, <b>Parag Jain</b> , Ayushi Dalmia, Abhishek Bansal, Ashish Mittal, Karthik Sankaranarayanan<br>Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)                                                                                             | 2019 |
| <b>Unsupervised Neural Text Simplification</b><br>Sai Surya, Abhijit Mishra, Anirban Laha, <b>Parag Jain</b> , Karthik Sankaranarayanan<br>Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)                                                                                                                            | 2018 |
| <b>A Mixed Hierarchical Attention Based Encoder-Decoder Approach for Standard Table Summarization</b><br><b>Parag Jain</b> , Anirban Laha, Karthik Sankaranarayanan, Preksha Nema, Mitesh M. Khapra, Shreyas Shetty<br>Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)2018                |      |
| <b>Generating Descriptions from Structured Data Using a Bifocal Attention Mechanism and Gated Orthogonalization</b><br>Preksha Nema, Shreyas Shetty, <b>Parag Jain</b> , Anirban Laha, Karthik Sankaranarayanan, Mitesh M. Khapra<br>Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) 2018 |      |
| <b>Story Generation from Sequence of Independent Short Descriptions</b><br><b>Parag Jain</b> , Priyanka Agrawal, Abhijit Mishra, Mohak Sukhwani, Anirban Laha, Karthik Sankaranarayanan<br>Special Interest Group on Knowledge Discovery in Data (KDD)                                                                                                              | 2017 |
| <b>Creation and Interaction with Large-Scale Domain-Specific Knowledge Bases</b><br>Shreyas Bharadwaj et al. [and others, including <b>Parag Jain</b> ]<br>Proceedings of the Very Large Data Bases Endowment (VLDB)                                                                                                                                                | 2017 |
| <b>Online Active Metric Learning for Clustering</b><br><b>Parag Jain</b> , Vineeth N. Balasubramanian<br>Research Colloquium in the 3rd Xerox Research Centre India Open (XRCI Open)                                                                                                                                                                                | 2016 |
| <b>Patents</b>                                                                                                                                                                                                                                                                                                                                                      |      |
| <b>Cognitive assistant for co-generating creative content</b> – US Patent US20200134089A1<br>Anush Sankaran, Pranay Lohia, Priyanka Agrawal, Disha Shrivastava, Anirban Laha, Parag Jain                                                                                                                                                                            | 2022 |

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| <b>Real-time assessment of text consistency</b> – US Patent US11194964B2<br>Abhijit Mishra, Anirban Laha, Parag Jain, Karthik Sankaranarayanan                                             | 2021 |
| <b>Controllable Style-Based Text Transformation</b> – US Patent US11734509B2<br>Abhijit Mishra, Parag Jain, Amar P. Azad, Karthik Sankaranarayanan                                         | 2021 |
| <b>Unsupervised Tunable Stylized Text Transformations</b> – US Patent US11574132B2<br>Parag Jain, Amar P. Azad, Abhijit Mishra, Karthik Sankaranarayanan                                   | 2021 |
| <b>Structured article generation</b> – US Patent US20210117431A1<br>Parag Jain, Abhijit Mishra, Anirban Laha, Saravanan Krishnan, Karthik Sankaranarayanan                                 | 2021 |
| <b>Script modification</b> – US Patent US20200125638A1<br>Saneem Chemmengath, Parag Jain, Anirban Laha, Saravanan Krishnan                                                                 | 2021 |
| <b>Ontology-based automatic bootstrapping of state-based dialog systems</b> – US Patent US20190228068A1<br>Jaydeep Sen, Parag Jain, Diptikalyan Saha, Ashish Mittal                        | 2020 |
| <b>Generation of variable natural language descriptions from structured data</b> – US Patent US20200073944A1<br>Abhijit Mishra, Parag Jain, Anirban Laha, Karthik Sankaranarayanan         | 2020 |
| <b>Displaying dynamic content on multiple devices</b> – US Patent US10664217B1<br>Anirban Laha, Vijay Ekambaram, Parag Jain, Disha Shrivastava                                             | 2020 |
| <b>Simulating tactile information for haptic technology</b> – US Patent US20190163272A1<br>Shreya Khare, Parag Jain, Srikanth G. Tamilselvam, Senthil Kumar Kumarasamy Mani, Sampath Dechu | 2019 |

## Invited Talks

- Generating and Reasoning over Structured Data Meta, Amazon, 2024
- Structsum Generation using Large Language Models IIT Hyderabad, 2024
- Foster Talk at Jabalpur Engineering College 2023
- Conversational Semantic Parsing - Online International Conference On Advances in Physical, Mathematical and Computational Sciences, India 2022
- Introduction to Machine Learning - Christ University Bangalore 2018
- Metric Learning for Clustering in Streaming Large-Scale Data - IBM Bangalore 2016
- Introduction to PageRank, IIT Hyderabad 2015

## Awards

- Travel Award by Google DeepMind to present our work at ACL 2024
- Travel Award by Facebook AI Research to present our work at EurNLP 2019
- PhD Scholarship Award by Huawei 2019
- IBM Second Invention Plateau Award - 2019
- IBM First Invention Plateau Award - 2019
- Equity Award for outstanding achievement in research and development. - 2018  
(An individual award selected globally every year by the office of CEO of IBM.)
- IBM Research First Patent Award - 2018
- Certificate of Appreciation from the Director of IBM Research India for my Contribution to Natural Language Generation Research in 2018
- Travel Grant by Xerox Research for attending XRCI Open, Bangalore - 2016
- Finalist Amazon Idea hackathon, Hyderabad 2016
- Travel Grant for attending the Fifth User Interaction Workshop at INCOIS, Hyderabad - 2015
- Ministry of Human Resource Development Scholarship by Government of India, 2013 - 2015
- Ministry of Human Resource Development Scholarship by Government of India, 2012

## Service

- Actively contribute as a reviewer for leading machine learning and NLP conferences such as COLM, ACL, EMNLP, and NAACL.
- Founding team member of TEDxIITHHyderabad, 2016
- Local coordinator for i.school workshop University of Tokyo, at IIT Hyderabad, 2015
- Student placement representative CSE Department IIT Hyderabad, 2015
- Core member of organizing team Aureoleâ€™10 Engineering Techfest, 2010